

KILN

Operations & Quality

The execution, quality and uptime specialist. Explains what the plant is making, whether it passes inspection, and whether the machines will still be there tomorrow.

Who this is for: product, engineering, operations and anyone who has to explain XELOR to somebody else.

What it covers: the work KILN reasons about, the rules it cannot break, the exact tools it can call, and what is honestly not built yet.

Truth standard: the repository as it stands on 8 August 2026.

KILN

What KILN is for

A role with a fixed tool list, not a general assistant

In one sentence

Explains what the plant is making, whether it passes inspection, and whether the machines will still be there tomorrow.

3

business areas it speaks for

8

tools it can actually call

4

capability families allowed

0

agents it may delegate to

What reaches it

What the plan says to make · Whether components can be issued · Readings, verdicts and dispositions · Downtime and preventive-maintenance signals · Stored robot, AMR and material-dwell records from Factory Connect

What it hands back

Whether the order can finish · Where the quality evidence is thin · Reviewable audit and CAPA drafts · An operations work item

Capability families it is allowed to touch

production.

quality.

maintenance.

agent.

Ownership and authority are not the same thing

The areas above are where KILN is the right specialist to ask. The tool table on page 8 is the much smaller set it can invoke at runtime. Owning a business area does not grant runtime power over it — and for ONYX, the supervisor, the gap between the two is the widest in the system.

What sits behind KILN

Records, screens, workflow, controls and hand-offs

Production KILN module · production	
Purpose	Executes a manufactured order through a pinned material list and ordered operations, then moves finished quantity into stock.
Core records	Production orders, BOM/component snapshot, operation sequence/status, operator/evidence notes, issue movements, completion quantity and finished-goods receipt references.
User surface	Machines & robot cells with an operational map; Production orders; Order detail with component issue, operation progress and completion actions.
End-to-end flow	Create from a manufactured item/BOM; snapshot components; define/start/complete operations in predecessor order; issue components through Inventory; require all routed work complete; receive finished goods; expose the full execution trail.
Enforced controls	Idempotency, predecessor gates, accountable operator/evidence, pinned BOM, all-blockers-at-once completion check, tenant RLS and atomic inventory movements.
Inputs and outputs	Receives make proposals from Planning, consumes Engineering BOM and Inventory components, joins Factory Connect robot/cell evidence, exposes lots to Quality, and reports execution context to Maintenance and Decision Commander.
Current boundary	Order, operation, issue and receipt workflows are live. The robot-cell screen shows stored demo/API-submitted records; no physical controller or plant protocol adapter is connected. The quality completion gate, scrap posting and detailed genealogy/costing remain incomplete.
Primary code map	apps/web/src/modules/production/manifest.ts · apps/api/src/modules/production/

QMS & Audit KILN module · quality	
Purpose	Records inspection evidence and turns defects into controlled findings, corrective actions, training and audit-ready evidence packs.
Core records	Specifications and revisions, sampling plans/snapshots, inspections/readings/verdicts, dispositions, findings/NCRs, CAPAs/actions/effectiveness reviews, documents, training, audits and evidence packs.
User surface	Overview; Inspections; Documents; Audits; Findings; Corrective actions; Training; Evidence packs; Inspection detail.
End-to-end flow	Select a sampling plan from lot size; snapshot the plan and applied limits; record readings; calculate pass/reject deterministically; quarantine/release through Inventory; open a finding; investigate cause and actions; let a person judge effectiveness; collect and version audit evidence.
Enforced controls	Immutable applied limits, critical-defect override, sample derivation, disposition/movement linkage, no-delete history, human-owned CAPA effectiveness and draft-only agent outputs.
Inputs and outputs	Consumes Production or receipt context, moves held stock through Inventory, informs Service complaints, and supplies KILN/HEXA with traceable evidence for QMS missions.
Current boundary	Inspection and NCR/CAPA depth are live. Automatic production/receipt inspection creation, calibration/gauge management and complete material genealogy are missing.
Primary code map	apps/web/src/modules/quality/manifest.ts · apps/api/src/modules/quality/

How to read these module pages

“Live” means the records, service rules and user/API surface exist in this repository. It does not mean every surrounding enterprise process is complete. The boundary row names what remains illustrative, manual, simulate-driven or outside the MVP.

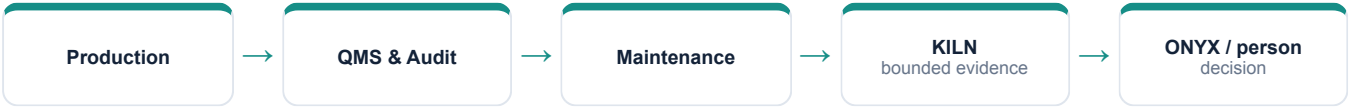
How KILN's modules connect

A module owns its records; the agent explains the combined evidence

Maintenance		KILN module · maintenance
Purpose	Keeps assets safe and available through requests, controlled work, downtime history, preventive schedules and reproducible reliability measures.	
Core records	Assets/hierarchy/work-centre links, readings, requests, work orders, labour/tasks/safety flags/spares/external work, downtime intervals/corrections/disputes, PM schedules/occurrences and KPI snapshots.	
User surface	Assets; Asset detail; Work orders; Requests; Downtime; Preventive maintenance; KPIs.	
End-to-end flow	Register an asset; receive/acknowledge/triage a floor request; assign and execute a work order with tasks, labour and spares; record asset downtime from stop to handback; close only after gates pass; generate PM work and calculate reliability from source intervals.	
Enforced controls	Database exclusion prevents overlapping downtime, corrections retain original timestamps, completion reports every missing condition, spare issue uses Inventory, and KPI snapshots are marked stale after relevant corrections.	
Inputs and outputs	Links assets to Production work centres, uses Inventory spares, records external-work purchasing context and feeds KILN/Decision Commander with uptime risk.	
Current boundary	Core maintenance workflow and KPIs are live. Factory Connect can store bounded operational-event submissions, but does not authenticate them as sensor/controller-originated evidence; predictive models, certified adapters, shift calendars and automatic external procurement are not complete.	
Primary code map	apps/web/src/modules/maintenance/manifest.ts · apps/api/src/modules/maintenance/	

Cross-module coordination

Production records the work, Quality decides whether the result is acceptable, and Maintenance explains whether the equipment can sustain the plan. Inventory movements are shared evidence, while quality and downtime conclusions remain deterministic and human-accountable.



How KILN's world actually runs

The business pipeline, not the agent plumbing

Three cycles that share one rule: the record is made where the work happens, and the arithmetic that judges it is code, not a model. No AI feature is registered for Quality or Maintenance, deliberately.

1

Make — explode once

At order creation the BOM is exploded into a component list and pinned. A later BOM revision cannot rewrite a job already on the floor.

2

Make — issue and operate

Components leave stores through Inventory's write path. Operations run in sequence: one cannot start while its predecessor is open, and completing one needs an accountable operator and an evidence note.

3

Make — complete

Finished goods are received into stock. A production order cannot complete while any routed operation is still open.

4

Check — sample

Lot size selects a sample from a plan table. The derivation is stored in words on the inspection, so a later reader sees why eight and not five.

5

Check — judge

Limits are SNAPSHOTTED onto each reading. A later spec revision must never silently change the verdict of a historical inspection. One critical failure rejects the lot regardless of the accept number.

6

Check — dispose

A rejected lot moves warehouse to warehouse through the same stock path, and 'executed' without a movement reference is unrepresentable.

7

Keep running

Downtime measures the ASSET, not the paperwork — the clock starts when the line stops, often before anyone triages it, and stops at handback rather than at closure.

How much of this KILN can actually see

Of everything above, KILN can read exactly 7 things directly — robot, amr, gateway and material-dwell evidence, production orders, inspection evidence, evidence view, audit checklist, investigation draft, evidence manifest. The rest of this pipeline is context for judging what it reads; it is not data the agent can reach. Where a mission needs more, it asks the specialist who owns it or it says the evidence is missing.

What the code will not let anyone do

Enforced by constraints and locks, not by wording

Overlapping downtime is impossible, not discouraged

A database exclusion constraint is the arbiter. The service's job on a collision is not to prevent it but to translate it into "join the existing interval".

Downtime is corrected, never deleted

The original timestamps are retained through every correction, and affected KPI snapshots are flagged stale rather than silently recomputed.

Zero failures means null, not zero

MTBF and MTTR are null when nothing broke, and with no shift calendar availability is null too — 24×7 is never assumed.

The completion gate reports every reason at once

A blocked work order returns all unmet conditions together, so a technician does not discover them one at a time.

Why this page matters more than the agent pages

An agent is only as trustworthy as the system it reads. Every rule here holds whether the request came from a person, a screen, a seeder or KILN — which is precisely why an agent can be given a read tool without also being given a way to do damage.

Where these rules are kept

apps/api/src/modules/production/production.service.ts

packages/platform/src/quality/sampling.ts

packages/platform/src/maintenance/reliability.ts

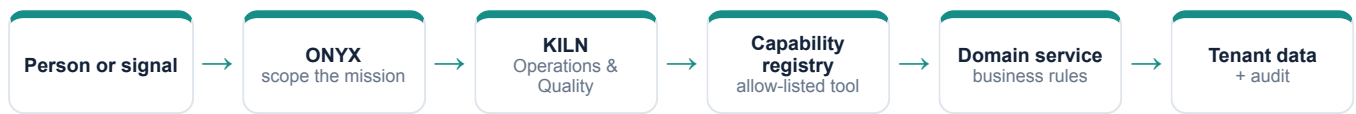
packages/platform/src/factory-connect/contracts.ts

apps/api/src/modules/integration/factory-connect.service.ts

apps/edge/src/runtime.ts

How KILN takes part in a mission

The engine controls order, parallelism and recovery



1 · Scope

ONYX turns the goal into a run against a frozen graph version, with a fixed tenant, user, step budget and timeout.

2 · Authorise

Two checks, both required: the tool must be on KILN's allow-list, and the signed-in person must independently hold the permission.

3 · Execute

A registered domain service does the work under the same rules a screen would face. There is no general SQL doorway.

4 · Preserve

Inputs, outputs, events and a checkpoint after every wave, so the run can be explained or resumed.

An agent can narrow authority, never widen it

KILN is a specialist and delegates to nobody. Because the permission check is repeated against the requesting person, a mission can never reach data that person could not have opened themselves.

Everything KILN can call

This table is the complete list — there is no other door

Capability	Mode	What comes back	Permission required	Needs approval
production.factory-connect.read	Read	Robot, AMR, gateway and material-dwell evidence	factory.connect.read	No
production.orders.read	Read	Production orders	production.order.read	No
quality.inspections.read	Read	Inspection evidence	quality.inspection.read	No
quality.evidence.collect	Analyse	Evidence view	quality.inspection.read	No
quality.audit-plan.draft	Draft	Audit checklist	quality.inspection.read	No
quality.capa-plan.draft	Draft	Investigation draft	quality.inspection.read	No
quality.audit-pack.draft	Draft	Evidence manifest	quality.inspection.read	No
agent.action.dispatch	Execute	Governed work item	agentos.run.operate	Yes

Read · Analyse · Simulate

Return evidence or a reproducible view. Nothing in the business changes.

Draft

Produce reviewable content that is labelled a draft and can never present itself as an approved outcome.

Execute

The only side-effecting mode in the whole system, and the only one that requires an approved human gate above it.

Both locks must open

The agent's allow-list AND the person's permission. Either one failing is a refusal.

What “dispatch” does and does not do

It appends a governed work item naming the responsible specialist, the approval it came from and the exact payload. It does not change a sales order, a stock balance, a production order, a journal or a customer message — a person still does that work.

Where KILN actually appears

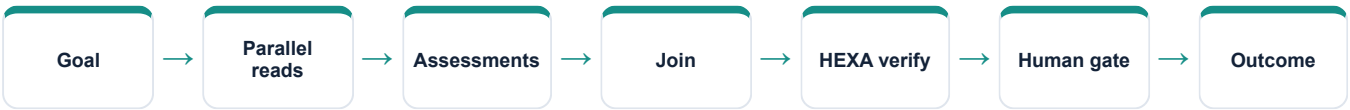
Node by node, from the registered graph definitions

Mission graph	What KILN does in it
Factory flow recovery	Reads stored robot and dwell evidence and explains recovery constraints; it neither determines local safety nor dispatches a machine command
Cross-functional readiness	Not involved
Nine-agent operating review	Reads production, then the operations assessment
Controlled action mission	Reads, recommends, and dispatches the execution priority
Working Capital Review	Not involved
QMS & Audit Readiness	Owns four of eight nodes — read, collect, draft the pack, explain the gaps
Managed Service Assurance	Not involved; physical asset restoration remains KILN work

Reading this honestly

“Not involved” is not a limitation, it is the design. A mission asks only the specialists whose evidence it needs, and a specialist cannot add itself to a graph at runtime. The graph is written down, versioned and content-hashed before the run starts.

The shape every mission shares



The twelve that did not pass

Real records from the running demo, not an illustration

Three work orders in the order the metal moved: 45 impellers, 45 shafts, then 40 finished pumps from the received heat.

At final inspection eight units were sampled. One measured 0.034 mm of shaft runout at the seal face against a 0.020 mm limit. Runout is classed CRITICAL here — a 316L process pump that weeps is a customer incident, not a quality one — so the lot was rejected.

Twelve units from that machining setup went to quarantine. Finished goods shows 28, not 40, and there is a ledger entry saying exactly where the other twelve went. Separately, Furnace 02 — the plant's only annealing route for 316L — lost vacuum mid-build: 4.5 hours, ₹1,855, cause recorded as seal wear found by the operator.

The sentence to say out loud

“KILN contributes evidence from its own area, with the source records behind it and its limits stated. It does not claim an action is done until the system has recorded and verified it.”

Fair questions to ask while watching

- Which records is this answer standing on?
- Which permission was checked, and against whom?
- Is this a recorded fact, a simulation, a draft, or a completed result?
- Would this step have waited for a person?
- What happens if the service restarts halfway through?

Live today, and deliberately not

The second column is the one worth reading

Working now

- Operations with predecessor gating and accountable evidence
- Snapshotted sampling plans and spec limits
- Critical-defect override on the lot verdict
- NCR → CAPA with a human-owned effectiveness decision
- Downtime, preventive schedules and reproducible KPIs
- Stored Factory Connect demo evidence for robot cells, AMRs, gateways and material dwell
- A governed factory-flow recovery mission with HEXA verification and a production-supervisor gate; the separate API result is a simulator policy evaluation only

Not built — stated plainly

- The manufacturing quality gate is built but unarmed — nothing creates the inspection it reads, so it never fires as shipped
- Purchase does not wire the receipt-side inspection gate either
- Rejected quantity is recorded on the operation but posts no scrap movement
- No calibration or gauge management, and no batch genealogy written from Production
- No physical robot, PLC or safety controller is connected; the standalone edge simulator is not wired to the Factory Connect API
- The ERP never replaces a robot controller, safety PLC, interlock, emergency stop or certified cell logic

Identity boundary

Verified token → tenant and actor

Authority boundary

Agent allow-list AND the person's permission

Action boundary

Side effects need an approved ancestor node

Data boundary

Domain service over a tenant-fenced database

Evidence boundary

Append-only runs, events, checkpoints and audit

Gaps are listed because a guide that hides them fails the first hour of technical due diligence. Every item on the right is either a roadmap decision or a known defect, and both are recorded in the project's own capability-gap register.

KILN on one page

For explaining the agent to somebody new

Its job

Explains what the plant is making, whether it passes inspection, and whether the machines will still be there tomorrow.

Speaks for

Production, Quality & QMS, Maintenance

Takes in

What the plan says to make · Whether components can be issued · Readings, verdicts and dispositions · Downtime and preventive-maintenance signals · Stored robot, AMR and material-dwell records from Factory Connect

Gives back

Whether the order can finish · Where the quality evidence is thin · Reviewable audit and CAPA drafts · An operations work item

Can call

production.factory-connect.read, production.orders.read, quality.inspections.read, quality.evidence.collect, quality.audit-plan.draft, quality.capa-plan.draft, quality.audit-pack.draft, agent.action.dispatch

Cannot do

The manufacturing quality gate is built but unarmed — nothing creates the inspection it reads, so it never fires as shipped · Purchase does not wire the receipt-side inspection gate either · Rejected quantity is recorded on the operation but posts no scrap movement

Words used on these pages

Agent	A named specialist with a fixed, allow-listed set of tools — not a process that can roam.
Capability	One registered operation with a declared mode, owner and required permission.
Mission graph	A versioned recipe: which steps run, which run together, where it waits for a person.
Checkpoint	A stored snapshot after each wave, used to explain a run and to resume it safely.
Governed work item	An approved, append-only assignment for a human team — not the business change itself.